

Introduction to Social Data Analytics: Week 2 Part 1

Today

- ▶ Introduce empirical module: Opportunity Insights data
- ▶ Introduce Stata
 - ▶ graphical user interface
 - ▶ Do-files
 - ▶ Basic commands for understanding your dataset

Intergenerational Mobility

- ▶ If you are born from low-income parents, what is the chance you move up the income distribution?
- ▶ Higher education is often viewed as a path for upward mobility
 - ▶ Differences in quality across colleges and rising costs of college may mean lower access/success for lower-income students
- ▶ Today we will introduce data made publicly available by **Opportunity Insights** that allows us to understand variation across colleges in the extent that they promote intergenerational mobility

Covid Didn't Kill Cities. Why Was That Prophecy So Alluring?

Why Record-Breaking Overnight Temperatures Are So Concerning

The Bond Market Is Telling Us to Worry About Growth, Not Inflation

A Planned Biden Order Aims to Tilt the Job Market Toward Workers

Birthday Parties as Virus Vector

For Surprise Medical Bills, It's the Beginning of the End

Biden Gained Will
Conservative Vote Shows

TheUpshot

FOLLOW US [f](#) [t](#) [in](#)
GET THE UPSHOT IN YOUR INBOX

SHARE

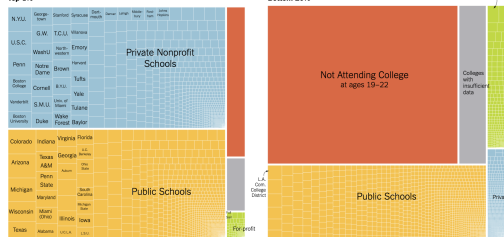
Some Colleges Have More Students From the Top 1 Percent Than the Bottom 60. Find Yours.

JAN. 18, 2017

RELATED ARTICLE

Where the top 1% and the bottom 20% go to college

Top 1%



Opportunity Insights Data

- ▶ The data comes from federal income tax returns
- ▶ Researchers at Opportunity Insights have linked to federal tax returns to tax records of universities and data from Pell grants
- ▶ Result is a dataset with:
 - ▶ Children linked to their parents (children are reported as dependents)

Opportunity Insights Data

- ▶ The data comes from federal income tax returns
- ▶ Researchers at Opportunity Insights have linked to federal tax returns to tax records of universities and data from Pell grants
- ▶ Result is a dataset with:
 - ▶ Children linked to their parents (children are reported as dependents)
 - ▶ Income data for both children and parents

Opportunity Insights Data

- ▶ The data comes from federal income tax returns
- ▶ Researchers at Opportunity Insights have linked to federal tax returns to tax records of universities and data from Pell grants
- ▶ Result is a dataset with:
 - ▶ Children linked to their parents (children are reported as dependents)
 - ▶ Income data for both children and parents
 - ▶ Children linked to the universities they attended
- ▶ This was a **huge** data task that gives us a new way to study an important question: how do colleges vary in the extent that they promote intergenerational mobility

Intergenerational Mobility by College

- ▶ To measure intergenerational mobility, we need to compare parent's income to income of their children
- ▶ Inflation (general rise in prices) will make this difficult to do by looking directly at income
- ▶ To measure intergenerational mobility, we will focus on **quantiles**

Quantiles

- ▶ Imagine we have 10 individuals and we sort them based on income (in the thousands per year)

$\{5, 6, 20, 29, 38, 42, 55, 62, 80, 88\}$

- ▶ We can break up the data into equal-sized groups based on income
- ▶ Each equal-sized group is a separate quantile

Example (Quintiles)

- Imagine we break the 10 individuals into 5 groups (or 5 quantiles)

$$\underbrace{\{5, 6\}}_{Q1}, \underbrace{\{20, 29\}}_{Q2}, \underbrace{\{38, 42\}}_{Q3}, \underbrace{\{55, 62\}}_{Q4}, \underbrace{\{80, 88\}}_{Q5}$$

- When we break up the observations into 5 equal-sized groups, this is referred to as breaking the data into **quintiles**
- Q1 is referred to as the first quintile (or lowest quintile)
- Q5 is referred to as the fifth quintile (or highest quintile)

Quintiles and Intergenerational mobility

- ▶ To measure intergenerational mobility, we need to compute the fraction of individuals coming from parents that have **low income** that are able to achieve **high income**
- ▶ The researchers at Opportunity Insights use quintiles to define **low income** and **high income**
- ▶ **Low income:** individuals with parents in the bottom quintile of earnings (Q1)
- ▶ **High income:** individuals that reach the top quintile of earnings (Q5)

College Mobility Rate

- ▶ Mobility rate for a college defined as

$$\text{mobility rate} = \underbrace{(\text{frac. of students from Q1})}_{\text{Access}} \times \underbrace{(\text{frac. of Q1 that reach Q5})}_{\text{Success}}$$

- ▶ Intuition: a college that admits a large fraction of low-income students (first term) and these students become high earners (second term) will promote intergenerational mobility
- ▶ This is the measure of mobility utilized by researchers at Opportunity Insights, but there are alternative measures that could be constructed

Getting Specific

- ▶ How do we (exactly) determine if a student comes from a low-income background?
- ▶ Take all parents that have a child in the same year (e.g. 1985)
- ▶ Compute average income for these parents when children are 15-19 (i.e. from 2000-2004)
- ▶ If parents' income is in bottom 20 percent over this period (i.e. in Q1), then classify students as coming from low-income background

Getting Specific

- ▶ How do we (exactly) determine if a student becomes a high earner?
- ▶ Take all individuals that are the same age and calculate earnings in 2014
- ▶ If individual's income is in the top 20 percent relative to individuals the same age (i.e. in Q5), then classify student as a high earner
- ▶ Now that we understand the variables we will be exploring, let's learn how to use Stata!

The Stata Graphical User Interface (GUI)

The screenshot displays the Stata/SE 16.1 graphical user interface. The top menu bar includes options like Open, Save, Log, Viewer, Graph, Do-file Editor, and Data Editor. The main window is divided into four panes:

- Results Window:** Displays the Stata logo, version 16.1, and copyright information for StataCorp. It also shows the Stata license: "Unlimited-user network, expiring 31 Jul 2021" and the serial number: "401609205573".
- Command Window:** A blank area for entering Stata commands.
- Variables Window:** A table with two columns: Name and Label. It is currently empty.
- Properties Window:** A table with two columns: Name and Value. It lists various properties for the current dataset, including Frame (default), Filename, Label, Notes, Variables (0), Observations (0), Size (0), Memory (64M), and Sorted by.

Large blue text labels are overlaid on the panes to identify them: "Results Window" over the Results pane, "Command Window" over the Command pane, "Variables Window" over the Variables pane, and "Properties Window" over the Properties pane.

Stata/SE 16.1

Open Save Log Viewer Graph Do-file Editor Data Editor More Break Search

Results

(R)

Statistics/Data analysis 16.1

Special Edition

Results Window

Copyright 1985–2019 StataCorp
StataCorp
4905 Lakeway Drive
College Station, Texas 77845
800–STATA–PC <https://www.stata.com>
979–696–4600 stata@stata.com
979–696–4601 (fax)

Stata license: Unlimited-user network, expiring 31 Jul 2021
Serial number: 401609205573

Command

Variables

Name	Label
------	-------

Properties

Variables	
Name	
Label	
Type	
Format	
Value label	
Notes	
Data	
Frame	default
File	
Filename	
Label	
Notes	
Variables	0
Observations	0
Size	0
Memory	64M
Sorted by	

Variables Window

Properties Window

/Users/Dave/Dropbox/Teaching/EP5/online/behavioral/llns/code

Command Window

- ▶ You can type in code through the command window
- ▶ Pressing enter will execute (or run) the code
- ▶ When we learn how to use Stata as a calculator, this is where we will type the code

Results Window

- ▶ When we execute code, any resulting output will be displayed in the Results window
- ▶ When we learn how to use Stata as a calculator, this is where the results will be displayed

Variables Window

- ▶ When we load data, the name of the variables in the dataset will appear in the first column
- ▶ Many times, datasets come with labels, that give us more description about the variable
- ▶ For example, in the figure below there is one variable named `income` with the label `Total Annual Income`

[illegible]

Properties Window

- ▶ This window gives us more information about the variables and data loaded in memory
- ▶ For example, in addition to the name and label of a variable, the type (i.e. numeric or string) and format of the variable (i.e. how the variable will be displayed)
- ▶ Additionally there is information about the size of the data (number of variables, observations, etc.)

Using Stata as a calculator

- ▶ To use Stata as a calculator you can just type `display` followed by a mathematical expression.
- ▶ For example, to compute $2+2$ type

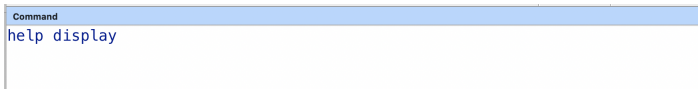
```
display 2+2
```

4

- ▶ The result will be visible in the Results window
- ▶ `display` is called a command in Stata. Stata comes with many commands that will be useful to explore and analyze our data

Help files

- ▶ Commands in Stata come with help files
- ▶ To access a help file type `help display` into the command window

A screenshot of the Stata Command Window. The window has a light blue header bar with the word "Command" in black text. Below the header, the text "help display" is entered in a blue monospaced font.

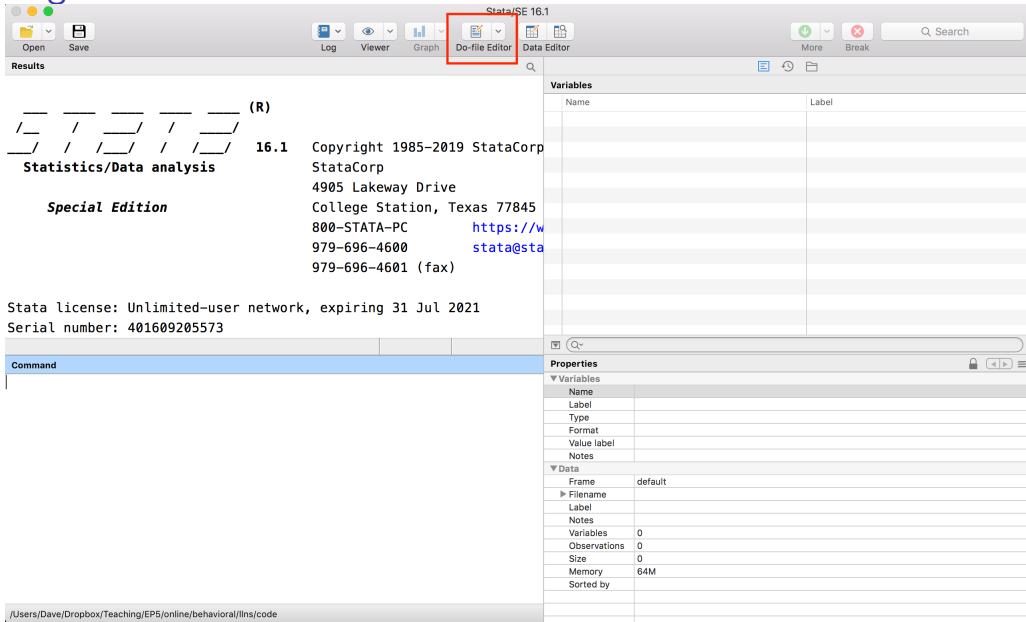
```
Command  
help display
```

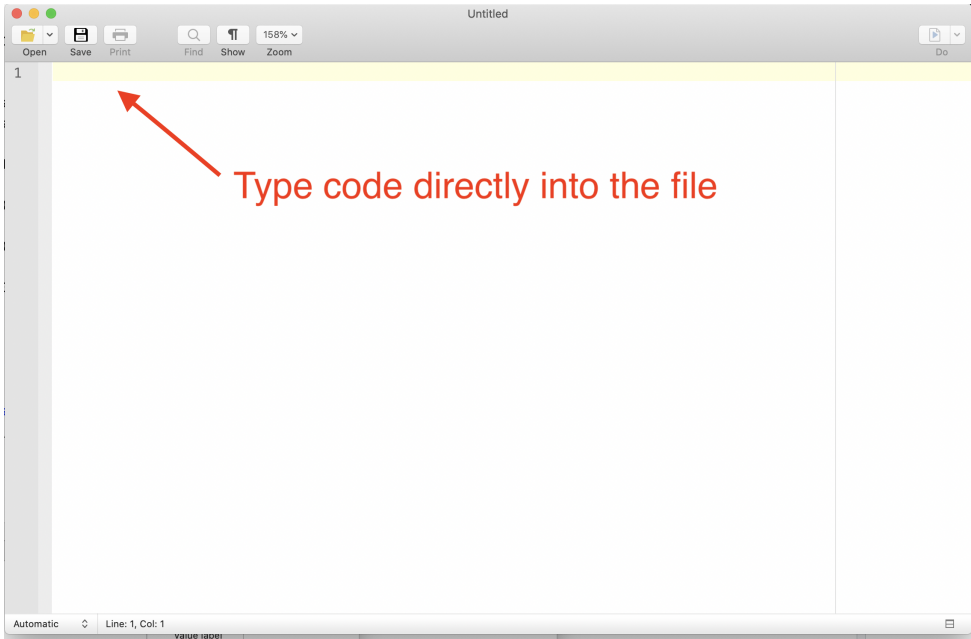
- ▶ This will open up a file that has instructions on how to use the given command

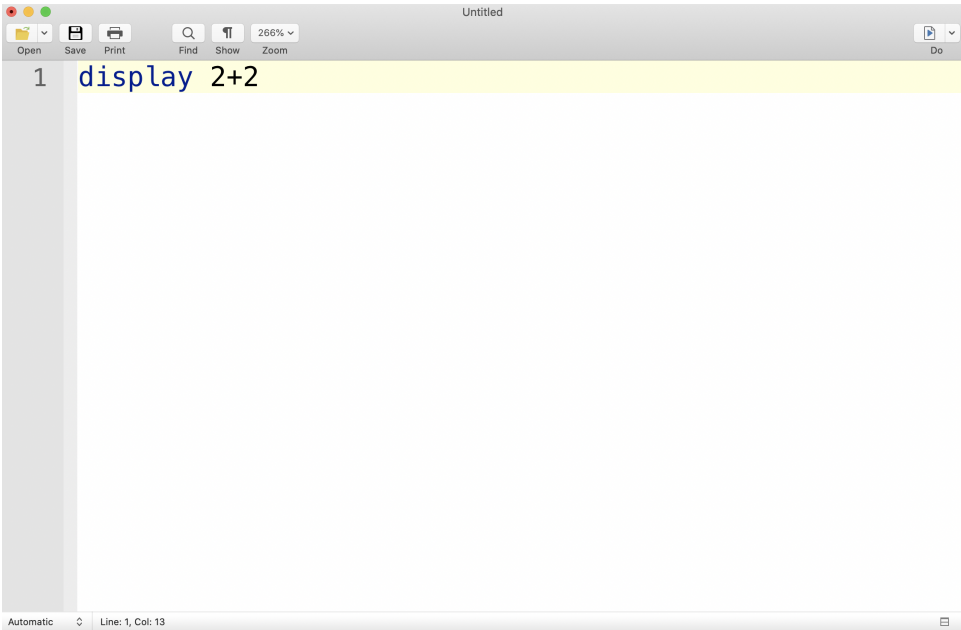
Syntax

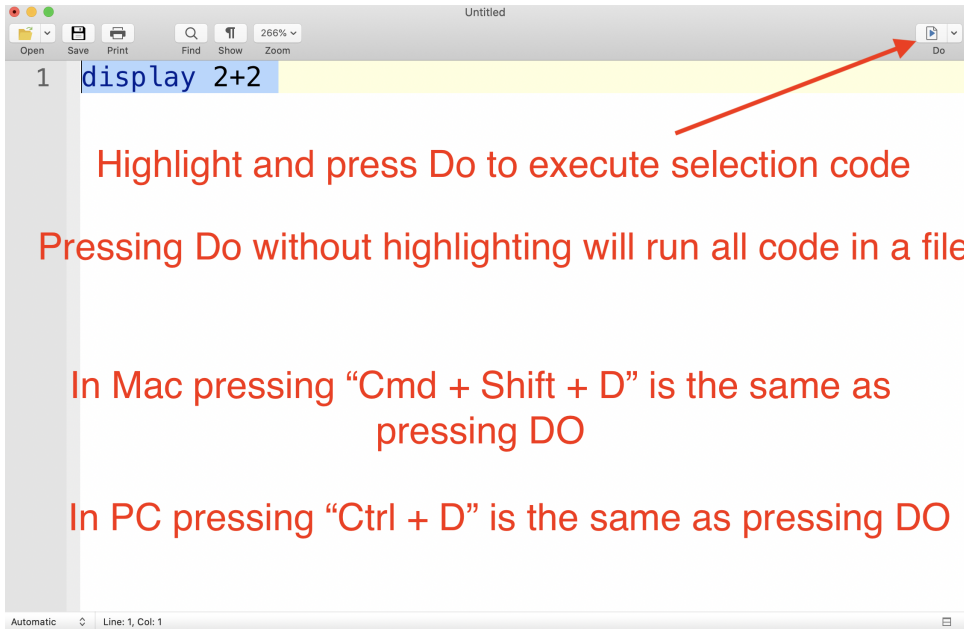
- ▶ Computer languages have different syntax
- ▶ Syntax refers to the format of the code
- ▶ It is similar to the “grammar” of the language
- ▶ If you use incorrect grammar in language, you may not be expressing the right meaning
- ▶ Similarly, if you use the wrong syntax in computer language, you may not get the correct output, and more than likely, an error will be reported

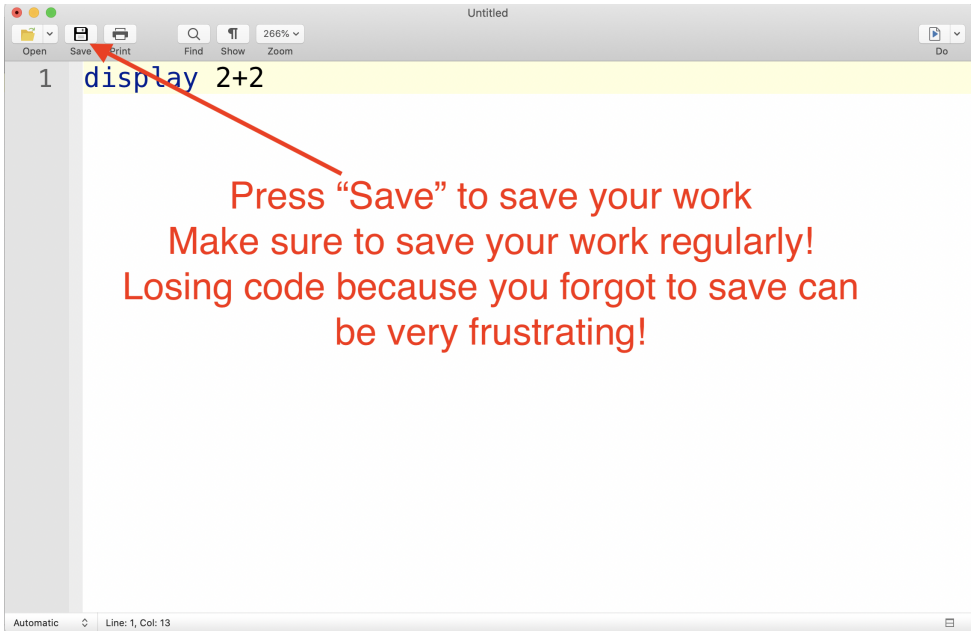
Opening a do-file

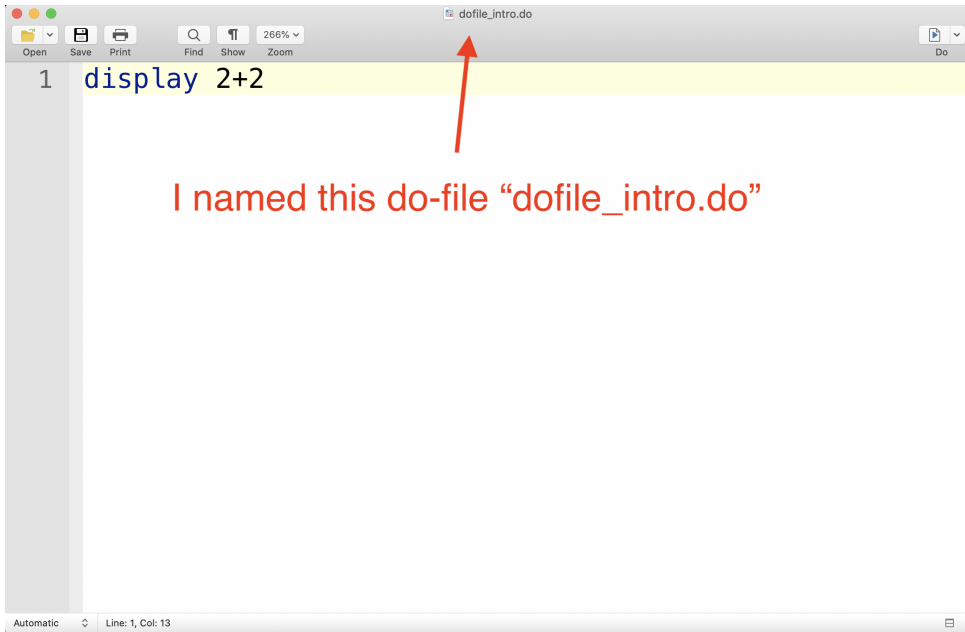


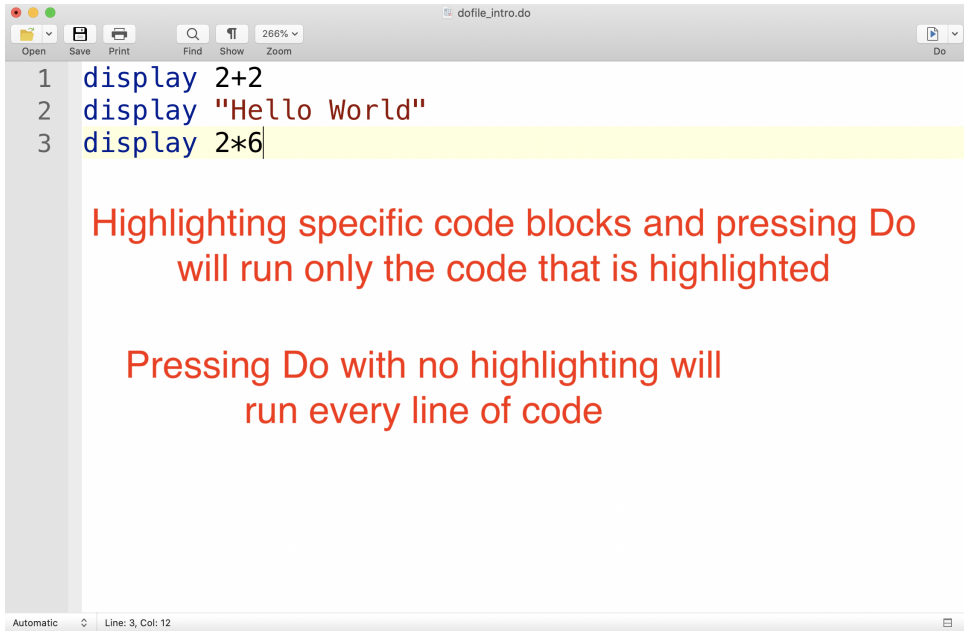


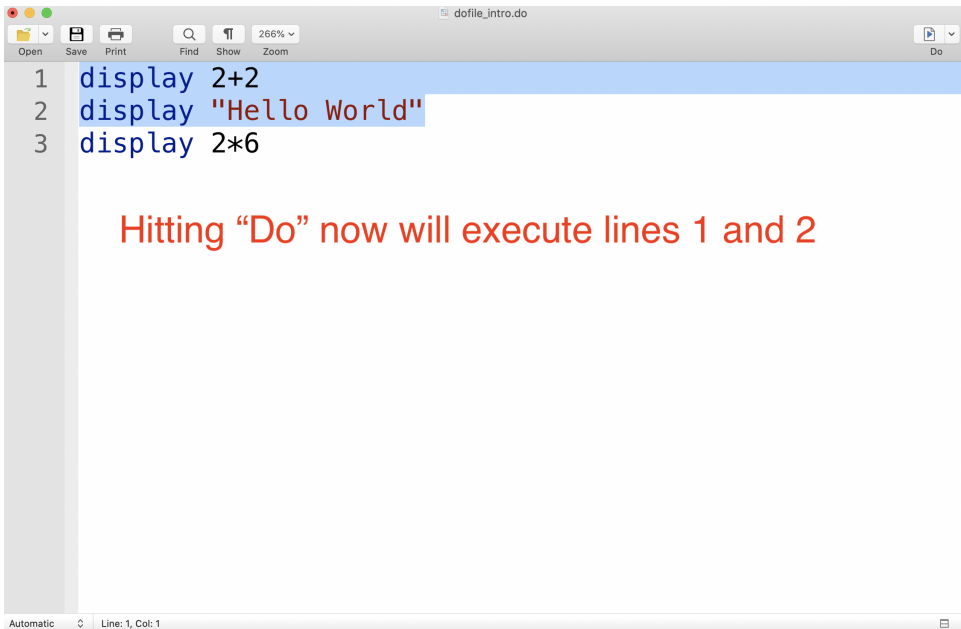












The Working Directory

- ▶ Every folder and file on your computer has an “address” associated with it
 - ▶ Referred to as a **file path**
- ▶ The **working directory** is the the default path used by a program
- ▶ Many commands in Stata (and R) will make use of the working directory

Where to set your working directory?

- ▶ In this class, the working directory should be set to where you save your data after you download it
- ▶ For me, the data is on `Dropbox/Teaching/EP5/online/02_week/videos/data`
- ▶ For you, I suggest making a folder somewhere convenient on your computer named something like EP5, and within that folder creates subfolders for each week of the course
 - ▶ Within that subfolder one of the folders should be named data
 - ▶ Whenever you download data for the week, place it in this folder

Change working directory

- ▶ To change your working directory type “cd” and then the path to folder you want to be your working directory

```
cd "/Users/davidarnold/Dropbox/Teaching/EP5/online/02_week/data"
```

- ▶ This is the path for me, but it will be different depending on where you put the downloaded data for the week

Loading Data

- ▶ To load data into Stata, we utilize the “use” command
- ▶ The general syntax is

```
use filename.dta, clear
```

- ▶ “filename.dta” should be replaced with whatever the name of the file is that you want to load
- ▶ “clear” tells Stata to clear anything that is currently loaded into memory
- ▶ To use this command “filename.dta” must be in the working directory!

```
cd "/Users/davidarnold/Dropbox/Teaching/EP5/online/02_week/data"  
use college_mobility.dta, clear
```

(Preferred Estimates of Access and Mobility Rates by College)

- ▶ Now the college mobility data is loaded into memory